Current: charge flow rate

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate current as charge transferred per time.

SUCCESS CRITERIA - I can...

1. Use I=q/t with seconds and coulombs.
2. Describe conventional current direction.
3. Explain steady charge conservation at a junction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

1. Keep charge separation fixed and multiply one charge magnitude by **3**.
2. **Compare electric-force magnitude**.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: I = Δq/Δt.
Current is charge per time; **1 A** = **1 C/s**. Conventional current follows positive charge motion.

Worked reasoning: If **12 C** pass a point in **4 s**, average current is I=Q/t=**3 A**. Charge circulates; a resistor does not consume charge.

Your turn: Draw a complete loop and **label equal current before** and after one resistor in steady state.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Draw conventional current around the** closed circuit, outside the battery from + toward -.
2. **What happens** if a wire is disconnected?

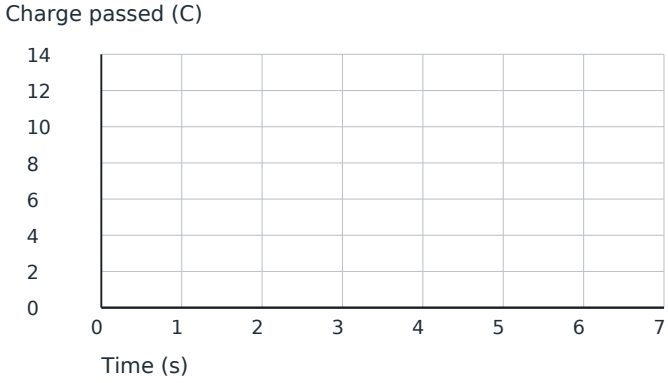


Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

04 Apply the physics

During a circuit test, **12 C** passes one point in a wire over **7 s**. Plot cumulative charge against time for a constant current. **Calculate current** from the graph's slope.



Marzano check - circle: ① ② ③

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05 Explain your reasoning

At a junction, **5 A** enters and **2 A** leaves along one branch.
1. **Find the other outgoing current** in steady operation.
2. **Explain the conservation rule**.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A siren emits **600 Hz** steadily while approaching a stationary listener. **Compare observed** and emitted frequency and explain.
Use wavefront spacing to **explain the frequency change**.

Source moves right. Add wavefronts.



Marzano check - circle: 1 2 3

07 Exit check

A current of **2 A** flows for **3 s**. **How much** charge passes?



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Construct two charge -time pairs that both give **2 A**. **Explain why different total charges** can give the same current.



Label component values, current and voltage as needed.

Marzano check - circle: 1 2 3